Brownian Resetting: A Renewal and First-Passage Atlas

HCS–C214 research certificate

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Abstract

We close a source-level theorem for one-dimensional Brownian motion with a Poisson reset to its starting point. The free process has an absolutely continuous Laplace stationary law, while the separately killed search process has an exact first-passage renewal transform. The mean hitting time, all Laplace moments, the unique positive reset optimum, and singular boundaries follow in one dimensionless calculation. The denominator is a renewal resolvent, not a dynamical zeta or a Fredholm determinant.

1 Two realizations

Fix $D, r, a > 0$. In the free realization on $\mathbb{R}$, $dX_t = \sqrt{2D}\, dW_t$ between independent rate-r resets to 0. In the search realization the same process starts at 0 and is killed at $T_a = \inf\{t \geq 0 : X_t = a\}$; its law on $(-\infty, a)$ is therefore sub-Markov after absorption. The stationary statement below belongs only to the free realization. The clock is physical elapsed time.

2 Free renewal theorem

Let

$$G_D(x,t) = \frac{1}{\sqrt{4\pi Dt}} \exp\left(-\frac{x^2}{4Dt}\right).$$

Conditioning on the last reset gives

$$p_r(x, t \mid 0) = e^{-rt} G_D(x,t) + r \int_0^t e^{-ru} G_D(x,u)\, du. \tag{1}$$

For $x \neq 0$, the integral equals

$$\frac{e^{-|x|\sqrt{r/D}}\, \mathrm{erfc}\big(\frac{|x|}{2\sqrt{Dt}} - \sqrt{rt}\big) - e^{|x|\sqrt{r/D}}\, \mathrm{erfc}\big(\frac{|x|}{2\sqrt{Dt}} + \sqrt{rt}\big)}{4\sqrt{Dr}},$$

and its continuous $x = 0$ value is $\mathrm{erf}(\sqrt{rt})/(2\sqrt{Dr})$. For every $t > 0$ this is an absolutely continuous Lebesgue density. Letting $t \to \infty$ gives the normalized free stationary law

$$p_{\mathrm{st}}(x) = \frac{1}{2}\sqrt{\frac{r}{D}}\, e^{-|x|\sqrt{r/D}}, \qquad \int_{\mathbb{R}} p_{\mathrm{st}}(x)\, dx = 1. \tag{2}$$

Equation (2) is not a stationary claim for the killed search.

A realization check

The distinction is operational, not cosmetic. At a fixed positive time the free law is a Lebesgue density: a reset at an exactly prescribed instant has probability zero, so the renewal integral does not hide a point mass. For the search process the state space is $(-\infty, a)$ with a Dirichlet boundary at a; probability that has reached the boundary is removed. Thus (2) can be used to describe long-time resetting on $\mathbb{R}$, but never to normalize the killed survival law.

3 Killed first passage

The no-reset transform from 0 to a is $f_0(q) = e^{-a\sqrt{q/D}}$. A first-reset decomposition yields

$$F_r(s) := \mathbb{E}[e^{-sT_a}] = \frac{(s+r)e^{-a\sqrt{(s+r)/D}}}{s + re^{-a\sqrt{(s+r)/D}}}, \qquad s \geq 0, \tag{3}$$

and the survival transform

$$S_r(s) := \int_0^\infty e^{-st}\,\Pr(T_a > t)\,dt = \frac{1 - e^{-a\sqrt{(s+r)/D}}}{s + re^{-a\sqrt{(s+r)/D}}}. \tag{4}$$

For $s > 0$, $1 - F_r(s) = sS_r(s)$; continuity at zero gives

$$\mathbb{E}[T_a] = -F_r'(0) = S_r(0) = \frac{e^{a\sqrt{r/D}} - 1}{r}.$$

All moments are finite for positive D, r, a. For every $n \geq 0$,

$$(-1)^n F_r^{(n)}(0) = \mathbb{E}[T_a^n], \qquad (-1)^n S_r^{(n)}(0) = \frac{\mathbb{E}[T_a^{n+1}]}{n+1}. \tag{5}$$

One direct way to see (3) is to write $f = f_0(s+r)$ for the no-reset transform during one exponentially distributed attempt. A reset before absorption contributes the factor $r(1-f)/(s+r)$ and starts an independent copy, hence

$$F_r(s) = f + \frac{r(1-f)}{s+r}F_r(s).$$

This scalar renewal equation also shows why its denominator is a resolvent of attempts rather than a periodic-orbit product.

4 Universal optimum and boundaries

Put $z = a\sqrt{r/D}$. Then

$$\frac{D}{a^2}\mathbb{E}[T_a] = g(z) := \frac{e^z - 1}{z^2}.$$

The derivative has the sign of $h(z) = z - 2(1 - e^{-z})$: $g(z) \to \infty$ at both endpoints, h has exactly one root in $(0, \infty)$ other than the limiting root at zero, and

$$z_* = 1.5936242600400400923\ldots, \qquad r_* = D(z_*/a)^2.$$

At $r = 0$ there is no normalizable stationary density and the half-line MFPT is infinite; at $a = 0$, $T_a = 0$; at $D = 0$, a reset path cannot reach a positive target. These are boundaries, not substitutions into Eq. (3).

5 Independent certificate and scope

The JSON certificate uses 81 propagator, 27 stationary, 9 normalization, 108 transform, and 27 MFPT rows at fixed rational sentinels. The checker recomputes the renewal integral by an independent $u = y^2$ quadrature and normalizes Eq. (2); SymPy checks the heat equation, renewal algebra, moment jet, and optimality derivative. Byte replay and hostile repaired-hash, stale-hash, and unknown-key tests are included.

As a second-pass audit, we also test the zero-reset, zero-target, and zero-diffusivity boundaries separately. This prevents a formal substitution of a singular boundary into the positive-parameter square-root formula and makes the theorem's scope machine-checkable.

realization	state space	quantity certified
free reset	$\mathbb{R}$	density and normalized Laplace stationary law
killed search	$(-\infty, a)$	survival mass and first-passage transform
boundary	$r = 0,\ a = 0,\ D = 0$	separate limiting behavior

The source formulas are attributed to Evans–Majumdar (2011); the finite certificate is a reproducibility artifact, not a priority claim. No numerical row selects z_*. It follows from the exact derivative of g, with the positive-root sign bracket independently checked at $z = 1$ and $z = 2$.

The source attribution is Evans–Majumdar (2011) and the resetting review by Evans–Majumdar–Schehr (2020); no priority or novelty is claimed. The strict Route-A verdict is

$$(\texttt{A0\_FAIL,A1\_FAIL,A2\_FAIL,A3\_FAIL,A4\_FAIL}), \quad \texttt{ROUTE\_A\_REJECTED}.$$

There is no intrinsic rational-prime carrier, primitive periodic-orbit clock, arithmetic divisor, or Hilbert–Polya operator. In particular, the denominator in Eq. (3) is never called a dynamical zeta.

References

[1] M. R. Evans and S. N. Majumdar, "Diffusion with stochastic resetting," *Phys. Rev. Lett.* 106, 160601 (2011), doi:10.1103/PhysRevLett.106.160601.

[2] M. R. Evans and S. N. Majumdar, "Diffusion with optimal resetting," *J. Phys. A* 44, 435001 (2011), doi:10.1088/1751-8113/44/43/435001.

[3] M. R. Evans, S. N. Majumdar and G. Schehr, "Stochastic resetting and applications," *J. Phys. A* 53, 193001 (2020), doi:10.1088/1751-8121/ab7cfe.